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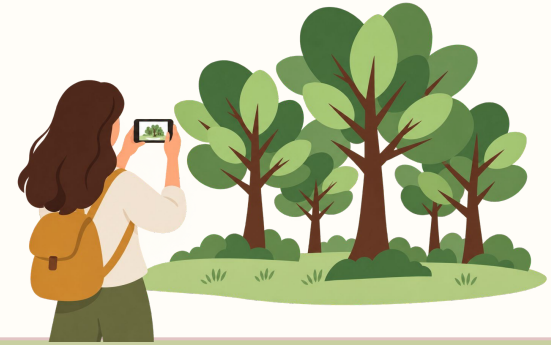
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PEI 26/27 - Group 2 - 29th September 2026



Context

Law No. 59/2021 requires municipalities to maintain an inventory of urban trees. Today, this survey is carried out manually, tree by tree.



Problems

- The process is slow;
- Spends too much resources;
- Gets outdated quickly.

GOALS

- 01** Make urban tree inventories faster and cheaper than manual field surveys;
- 02** Keep the inventory up to date with less effort;
- 03** Repeat surveys easily and track new or removed trees;
- 04** Scale to a whole city, not just a few streets;
- 05** Improve accuracy over time, instead of starting from scratch at each survey.



Expected Results:

- 01** Web application that automatically detects, locates and records trees from images and videos;
- 02** Georeferenced inventory without duplicates with an interactive map;
- 03** Manual validation, with corrections feeding model retraining;
- 04** Export, statistics by street or parish, and comparison between surveys;
- 05** Validation in a pilot area against a manual survey



TASKS

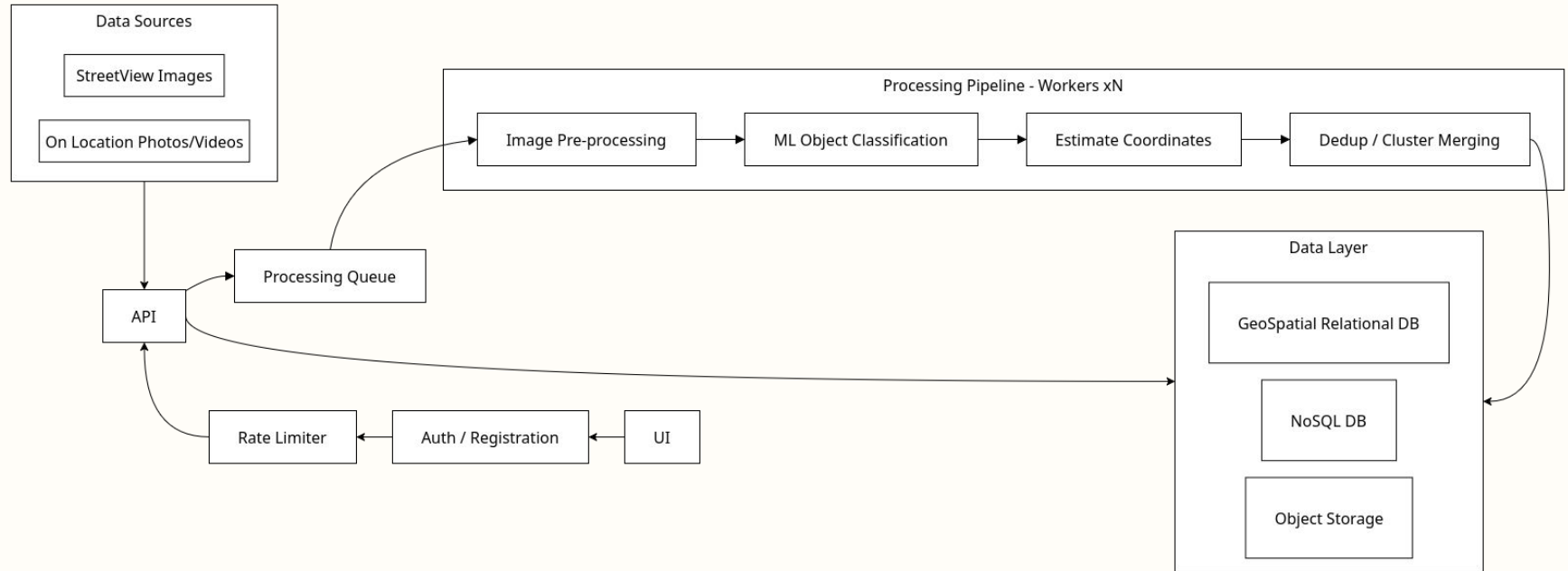
- 01 Analysis pipeline:** image and video ingestion, face and plate blurring, tree detection, triangulated positioning, and duplicate fusion
- 02 Web application:** interactive map, video upload, tree records, and manual validation
- 03 Data and infrastructure:** geospatial database, object storage, message queue, and CI
- 04 Model and data:** annotation, training, evaluation, and error analysis
- 05 Management and Documentation:** schedule, requirements, legal and risk analysis



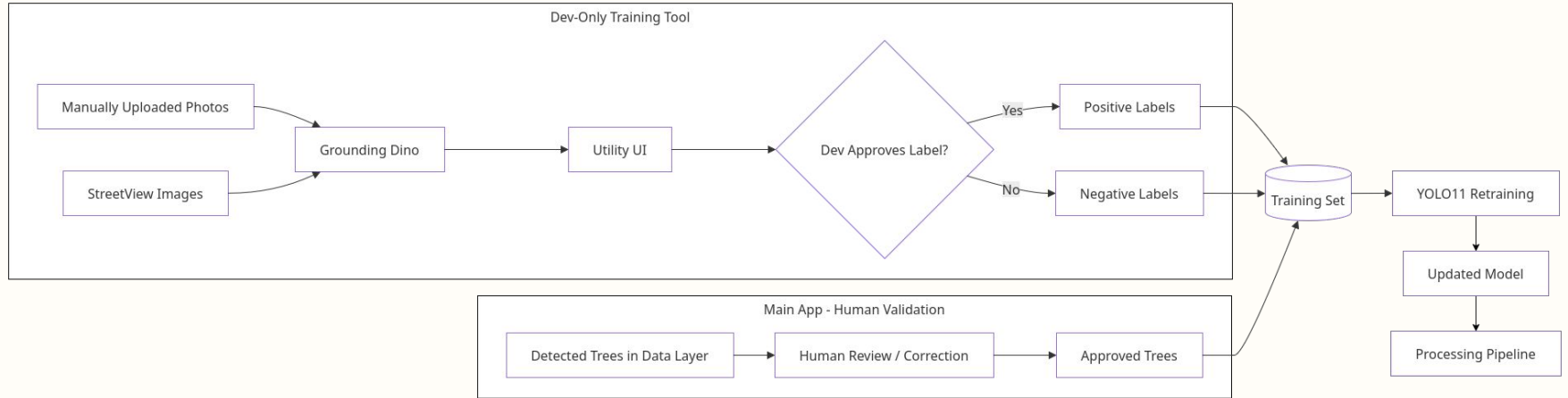
RELATED WORK

	Manual surveys	i-Tree Eco / Streets	OpenTree Map	iNaturalist	Research pipelines	TreeLens
Automatic detection	✗	✗	✗	✗	✓	✓
Individual tree geolocation	✗	✗	✓	✓	✓	✓
Easy to update and compare	✗	■	✗	■	✗	✓
No visit to each tree	✗	✗	✗	✗	✓	✓
Ready to use municipal tool	✓	■	■	✗	✗	✓

High-Level Architecture



Model's Retraining Process



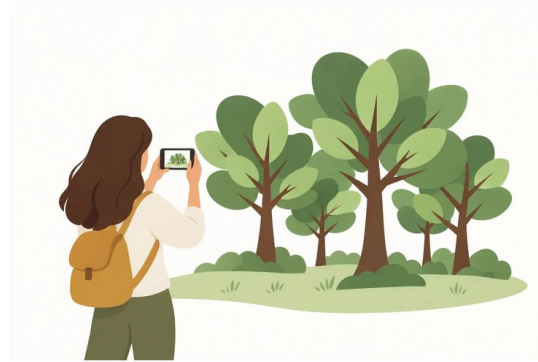
1st Semester Calendar



[Milestones](#)[Calendar](#)[Meetings](#)[GitHub](#)[Google Drive](#)[Jira](#)

Every Tree In Your City Detected, Located and Mapped

Because the urban tree inventory should not depend on tree-by-tree surveys

[Learn more](#)

Keep up with our work!

